

DMVPN Fase 3 con IKEv2 y OSPF

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Práctica: P4

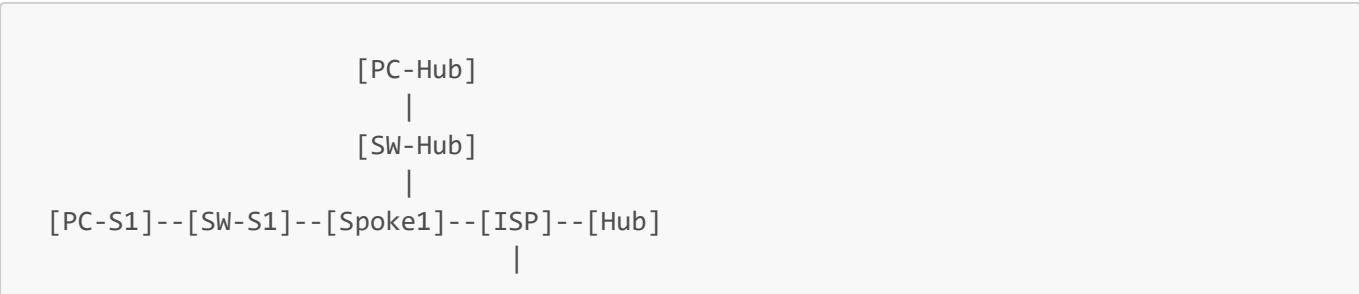
Objetivo

Configurar una VPN hub and spoke punto a multipunto **DMVPN (Dynamic Multipoint VPN) Fase 3** con IKEv2 y enrutamiento dinámico OSPF entre un Hub y dos Spokes (Cisco CSR1000v), permitiendo comunicación cifrada tanto entre Hub↔Spoke como directamente entre Spoke↔Spoke. Esta práctica combina la arquitectura DMVPN de P7 con el protocolo de negociación IKEv2, representando la implementación más moderna y escalable de DMVPN.

Diferencias entre Fase 2 (P7) y Fase 3 (P8)

Característica	Fase 2 (P7 — IKEv1)	Fase 3 (P8 — IKEv2)
Protocolo IKE	IKEv1	IKEv2
Mensajes de negociación	6-9 (Main Mode)	4 (IKE_SA_INIT + IKE_AUTH)
Comando de verificación	<code>show crypto isakmp sa</code>	<code>show crypto ikev2 sa</code>
Estado SA activa	<code>QM_IDLE</code>	<code>READY</code>
Túnel Spoke↔Spoke	Directo	Directo
Routing en Spokes	Ruta específica por Spoke	Ruta resummarizada via Hub (más escalable)
NHRP redirect	Sí (Hub)	Sí (Hub)
NHRP shortcut	Sí (Spokes)	Sí (Spokes)
Configuración IKE Hub	<code>crypto isakmp policy/key</code>	<code>crypto ikev2 proposal/policy/keyring/profile</code>
Escalabilidad	Media	Alta

Topología



[Spoke2]--[SW-S2]--[PC-S2]

Túneles DMVPN (mGRE sobre IPSec IKEv2):
Hub (10.0.0.1) ↔ Spoke1 (10.0.0.2)
Hub (10.0.0.1) ↔ Spoke2 (10.0.0.3)
Spoke1 (10.0.0.2) ↔ Spoke2 (10.0.0.3) [túnel directo Fase 3]

Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Máscara	Descripción
Hub	GigabitEthernet1	25.7.73.193	255.255.255.252	WAN Hub → ISP
ISP	GigabitEthernet1	25.7.73.194	255.255.255.252	WAN ISP → Hub
ISP	GigabitEthernet2	25.7.73.197	255.255.255.252	WAN ISP → Spoke1
ISP	GigabitEthernet3	25.7.73.201	255.255.255.252	WAN ISP → Spoke2
Spoke1	GigabitEthernet2	25.7.73.198	255.255.255.252	WAN Spoke1 → ISP
Spoke2	GigabitEthernet2	25.7.73.202	255.255.255.252	WAN Spoke2 → ISP
Hub	GigabitEthernet3	25.7.73.205	255.255.255.248	Gateway LAN Hub
Spoke1	GigabitEthernet3	25.7.73.213	255.255.255.248	Gateway LAN Spoke1
Spoke2	GigabitEthernet3	25.7.73.221	255.255.255.248	Gateway LAN Spoke2
Hub	Tunnel0 (mGRE)	10.0.0.1	255.255.255.0	NHS (Hub DMVPN)
Spoke1	Tunnel0 (mGRE)	10.0.0.2	255.255.255.0	NHC Spoke1
Spoke2	Tunnel0 (mGRE)	10.0.0.3	255.255.255.0	NHC Spoke2
PC-Hub	eth0	25.7.73.206	255.255.255.248	Host LAN Hub
PC-S1	eth0	25.7.73.214	255.255.255.248	Host LAN Spoke1
PC-S2	eth0	25.7.73.222	255.255.255.248	Host LAN Spoke2

Subredes utilizadas (bloque 25.7.73.192/27)

Subred	Rango utilizable	Uso
25.7.73.192/30	25.7.73.193 – 25.7.73.194	WAN Hub – ISP
25.7.73.196/30	25.7.73.197 – 25.7.73.198	WAN ISP – Spoke1
25.7.73.200/30	25.7.73.201 – 25.7.73.202	WAN ISP – Spoke2
25.7.73.204/29	25.7.73.205 – 25.7.73.210	LAN Hub
25.7.73.208/29	25.7.73.209 – 25.7.73.214	LAN Spoke1
25.7.73.216/29	25.7.73.217 – 25.7.73.222	LAN Spoke2

Subred	Rango utilizable	Uso
10.0.0.0/24	10.0.0.1 – 10.0.0.254	Red DMVPN (Tunnel0)

Parámetros de configuración

IKEv2 Proposal

Parámetro	Valor
Cifrado	AES-CBC 256
Integridad	SHA-256
Grupo DH	Group 14 (2048-bit)

IKEv2 Keyring

Parámetro	Valor
Peer	0.0.0.0/0.0.0.0 (cualquier)
Pre-shared key	Cisco123!

IKEv2 Profile

Parámetro	Valor
Match identity remote	0.0.0.0 (cualquier)
Autenticación local	pre-share
Autenticación remota	pre-share

IPSec Transform Set

Parámetro	Valor
Cifrado ESP	AES 256
Integridad ESP	SHA-256 HMAC
Modo	Transport

Tunnel mGRE

Parámetro	Hub	Spoke1	Spoke2
IP Tunnel	10.0.0.1/24	10.0.0.2/24	10.0.0.3/24
Tunnel source	Gi1	Gi2	Gi2
Tunnel mode	gre multipoint	gre multipoint	gre multipoint

Parámetro	Hub	Spoke1	Spoke2
NHRP network-id	100	100	100
NHRP NHS	—	10.0.0.1	10.0.0.1
ip nhrp redirect	Sí	No	No
ip nhrp shortcut	No	Sí	Sí
ip ospf network	broadcast	broadcast	broadcast
ip ospf priority	2 (DR)	0 (never DR)	0 (never DR)

Scripts de configuración

ISP

```
enable
configure terminal
hostname ISP

interface GigabitEthernet1
 ip address 25.7.73.194 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet2
 ip address 25.7.73.197 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.201 255.255.255.252
 no shutdown
 exit

end
write memory
```

Hub

```
enable
configure terminal
hostname Hub

interface GigabitEthernet1
 ip address 25.7.73.193 255.255.255.252
 no shutdown
 exit
```

```
interface GigabitEthernet3
 ip address 25.7.73.205 255.255.255.248
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 25.7.73.194
ip route 25.7.73.196 255.255.255.252 25.7.73.194
ip route 25.7.73.200 255.255.255.252 25.7.73.194

! --- IKEv2 Proposal ---
crypto ikev2 proposal PROP-DMVPN
 encryption aes-cbc-256
 integrity sha256
 group 14
 exit

! --- IKEv2 Policy ---
crypto ikev2 policy POL-DMVPN
 proposal PROP-DMVPN
 exit

! --- IKEv2 Keyring (accepta cualquier peer) ---
crypto ikev2 keyring KEY-DMVPN
 peer SPOKES
  address 0.0.0.0 0.0.0.0
  pre-shared-key Cisco123!
 exit
exit

! --- IKEv2 Profile ---
crypto ikev2 profile PROF-IKEv2-DMVPN
 match identity remote address 0.0.0.0
 authentication local pre-share
 authentication remote pre-share
 keyring local KEY-DMVPN
 exit

! --- Transform Set ---
crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
 mode transport
 exit

! --- IPSec Profile referenciando IKEv2 ---
crypto ipsec profile PROF-DMVPN
 set transform-set TS-DMVPN
 set ikev2-profile PROF-IKEv2-DMVPN
 exit

! --- Tunnel mGRE ---
interface Tunnel0
 ip address 10.0.0.1 255.255.255.0
 ip nhrp authentication Cisco123
 ip nhrp map multicast dynamic
```

```
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp redirect
ip ospf network broadcast
ip ospf priority 2
tunnel source GigabitEthernet1
tunnel mode gre multipoint
tunnel protection ipsec profile PROF-DMVPN
no shutdown
exit

! --- OSPF ---
router ospf 1
 network 10.0.0.0 0.0.0.255 area 0
 network 25.7.73.204 0.0.0.7 area 0
exit

end
write memory
```

Spoke1

```
enable
configure terminal
hostname Spoke1

interface GigabitEthernet2
 ip address 25.7.73.198 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.213 255.255.255.248
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 25.7.73.197
ip route 25.7.73.200 255.255.255.252 25.7.73.197
ip route 25.7.73.202 255.255.255.255 25.7.73.197

! --- IKEv2 Proposal ---
crypto ikev2 proposal PROP-DMVPN
 encryption aes-cbc-256
 integrity sha256
 group 14
 exit

! --- IKEv2 Policy ---
crypto ikev2 policy POL-DMVPN
 proposal PROP-DMVPN
 exit
```

```
! --- IKEv2 Keyring ---
crypto ikev2 keyring KEY-DMVPN
peer HUB
  address 0.0.0.0 0.0.0.0
  pre-shared-key Cisco123!
exit
exit

! --- IKEv2 Profile ---
crypto ikev2 profile PROF-IKEv2-DMVPN
match identity remote address 0.0.0.0
authentication local pre-share
authentication remote pre-share
keyring local KEY-DMVPN
exit

! --- Transform Set ---
crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
mode transport
exit

! --- IPSec Profile ---
crypto ipsec profile PROF-DMVPN
set transform-set TS-DMVPN
set ikev2-profile PROF-IKEv2-DMVPN
exit

! --- Tunnel mGRE ---
interface Tunnel0
ip address 10.0.0.2 255.255.255.0
ip nhrp authentication Cisco123
ip nhrp map 10.0.0.1 25.7.73.193
ip nhrp map multicast 25.7.73.193
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 10.0.0.1
ip nhrp shortcut
ip ospf network broadcast
ip ospf priority 0
tunnel source GigabitEthernet2
tunnel mode gre multipoint
tunnel protection ipsec profile PROF-DMVPN
no shutdown
exit

! --- OSPF ---
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 25.7.73.208 0.0.0.7 area 0
exit
```

```
end
write memory
```

Spoke2

```
enable
configure terminal
hostname Spoke2

interface GigabitEthernet2
 ip address 25.7.73.202 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.221 255.255.255.248
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 25.7.73.201
ip route 25.7.73.196 255.255.255.252 25.7.73.201
ip route 25.7.73.198 255.255.255.255 25.7.73.201

! --- IKEv2 Proposal ---
crypto ikev2 proposal PROP-DMVPN
 encryption aes-cbc-256
 integrity sha256
 group 14
 exit

! --- IKEv2 Policy ---
crypto ikev2 policy POL-DMVPN
 proposal PROP-DMVPN
 exit

! --- IKEv2 Keyring ---
crypto ikev2 keyring KEY-DMVPN
 peer HUB
  address 0.0.0.0 0.0.0.0
  pre-shared-key Cisco123!
 exit
exit

! --- IKEv2 Profile ---
crypto ikev2 profile PROF-IKEv2-DMVPN
 match identity remote address 0.0.0.0
 authentication local pre-share
 authentication remote pre-share
 keyring local KEY-DMVPN
 exit
```



```
! --- Transform Set ---
crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
mode transport
exit

! --- IPSec Profile ---
crypto ipsec profile PROF-DMVPN
set transform-set TS-DMVPN
set ikev2-profile PROF-IKEv2-DMVPN
exit

! --- Tunnel mGRE ---
interface Tunnel0
ip address 10.0.0.3 255.255.255.0
ip nhrp authentication Cisco123
ip nhrp map 10.0.0.1 25.7.73.193
ip nhrp map multicast 25.7.73.193
ip nhrp network-id 100
ip nhrp holdtime 300
ip nhrp nhs 10.0.0.1
ip nhrp shortcut
ip ospf network broadcast
ip ospf priority 0
tunnel source GigabitEthernet2
tunnel mode gre multipoint
tunnel protection ipsec profile PROF-DMVPN
no shutdown
exit

! --- OSPF ---
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 25.7.73.216 0.0.0.7 area 0
exit

end
write memory
```

SW-Hub, SW-S1, SW-S2 (IOSvL2)

```
enable
configure terminal

interface GigabitEthernet0/0
switchport mode access
switchport access vlan 1
no shutdown
exit

interface GigabitEthernet0/1
switchport mode access
```

```
switchport access vlan 1
no shutdown
exit

end
write memory
```

PC-Hub, PC-S1, PC-S2 (VPCS)

```
! PC-Hub
ip 25.7.73.206 255.255.255.248 25.7.73.205
save

! PC-S1
ip 25.7.73.214 255.255.255.248 25.7.73.213
save

! PC-S2
ip 25.7.73.222 255.255.255.248 25.7.73.221
save
```

Verificación y pruebas

Comandos ejecutados en Hub, Spoke1 y Spoke2.

Comandos de verificación

```
! En Hub
show dmvpn
show ip ospf neighbor
show crypto ikev2 sa
show ip route ospf

! En Spoke1 y Spoke2
show dmvpn
show crypto ikev2 sa
```

Resultados obtenidos — Hub

show dmvpn:

```
Type:Hub, NHRP Peers:2
25.7.73.198 10.0.0.2 UP 01:05:28 Dynamic ← Spoke1
25.7.73.202 10.0.0.3 UP 00:40:55 Dynamic ← Spoke2
```

show ip ospf neighbor:

```
25.7.73.213    0    FULL/DROTHER    Tunnel0 ← Spoke1
25.7.73.221    0    FULL/DROTHER    Tunnel0 ← Spoke2
```

show crypto ikev2 sa:

```
Tunnel-id  Local                Remote                Status
1           25.7.73.193/500        25.7.73.198/500      READY ← Hub↔Spoke1
2           25.7.73.193/500        25.7.73.202/500      READY ← Hub↔Spoke2
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, PSK
```

Resultados obtenidos — Spoke1

show dmvpn:

```
Type:Spoke, NHRP Peers:2
25.7.73.193    10.0.0.1    UP    Static    ← Hub
25.7.73.202    10.0.0.3    UP    Dynamic   ← Spoke2 (túnel directo Fase 3)
```

show crypto ikev2 sa:

```
Tunnel-id  Local                Remote                Status
1           25.7.73.198/500        25.7.73.193/500      READY ← Spoke1↔Hub
2           25.7.73.198/500        25.7.73.202/500      READY ← Spoke1↔Spoke2 directo
```

Pruebas de conectividad

Hub → Spokes:

```
Hub# ping 25.7.73.213
Success rate is 100 percent (5/5)
```

```
Hub# ping 25.7.73.221
Success rate is 100 percent (5/5)
```

Spoke1 → Spoke2:

```
Spoke1# ping 25.7.73.221 source GigabitEthernet3
! Primer intento: 0% (negociación NHRP + IKEv2)
```

```
! Segundo intento: 100% (túnel directo establecido)  
Success rate is 100 percent (5/5)
```

Spoke2 → Spoke1:

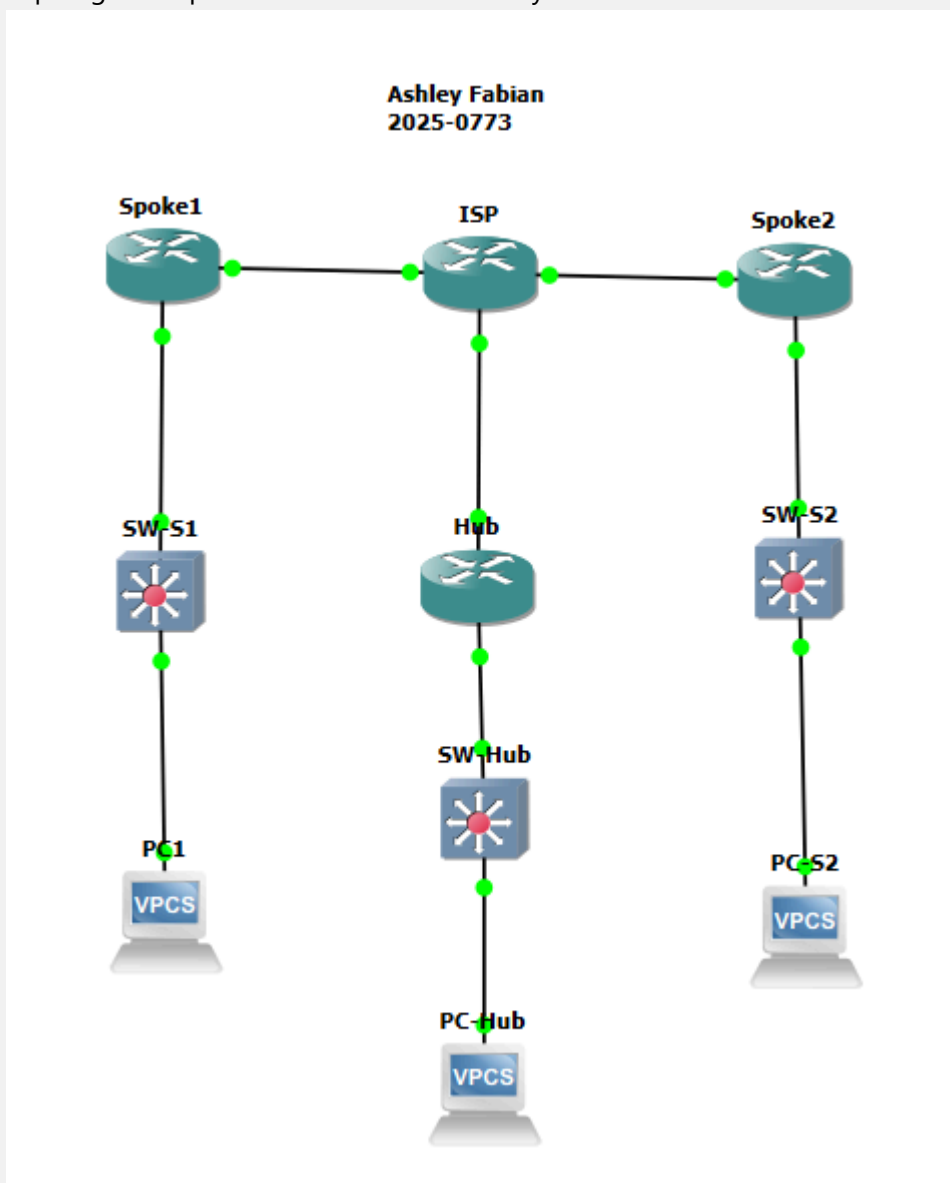
```
Spoke2# ping 25.7.73.213 source GigabitEthernet3  
Success rate is 100 percent (5/5)
```

El primer ping entre Spokes falla mientras NHRP resuelve la IP NBMA y se negocia la SA IKEv2 directa. Este es el comportamiento esperado y definitorio de DMVPN Fase 3. Una vez establecido el túnel directo, el tráfico entre Spokes ya no transita por el Hub.

Capturas de pantalla

Insertar aquí las capturas de:

1. Topología completa en GNS3 con nombre y matrícula visible



2. show dmvpn en Hub (2 peers UP Dynamic)

```

Hub#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
=====

```

```

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

```

#	Ent	Peer NBMA Addr	Peer Tunnel Add	State	UpDn Tm	Attrb
1		25.7.73.198	10.0.0.2	UP	01:12:57	D
1		25.7.73.202	10.0.0.3	UP	00:48:24	D

```
Hub#
```

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3. show ip ospf neighbor en Hub (2 vecinos FULL)

```
Hub#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
25.7.73.213	0	FULL/DROTHER	00:00:31	10.0.0.2	Tunnel0
25.7.73.221	0	FULL/DROTHER	00:00:39	10.0.0.3	Tunnel0

```
Hub#
```

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4. show crypto ikev2 sa en Hub (2 SAs READY)

```
Hub#show crypto ikev2 sa
IPv4 Crypto IKEv2 SA
```

Tunnel-id	Local	Remote	fvr/fivr	Status
1	25.7.73.193/500	25.7.73.198/500	none/none	READY

Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/4468 sec

Tunnel-id	Local	Remote	fvr/fivr	Status
2	25.7.73.193/500	25.7.73.202/500	none/none	READY

Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/2996 sec

```
IPv6 Crypto IKEv2 SA
```

```
Hub#
```

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5. show ip route ospf en Hub

```

Hub#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 25.7.73.194 to network 0.0.0.0

    25.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
O       25.7.73.208/29 [110/1001] via 10.0.0.2, 01:14:55, Tunnel0
O       25.7.73.216/29 [110/1001] via 10.0.0.3, 00:47:45, Tunnel0
Hub#

```

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6. show dmvpn en Spoke1 (peer dinámico hacia Spoke2)

```

Spoke1#
*Jun 29 00:21:14.248: %IOSXE-4-PLATFORM: R0/0: kernel: hrtimer: interrupt took 30108216 ns
Spoke1#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel

=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
  1 25.7.73.193          10.0.0.1  UP 01:15:46    S

Spoke1#

```

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7. show crypto ikev2 sa en Spoke1 (2 SAs READY — Hub y Spoke2)

```

Spoke1#show crypto ikev2 sa
IPv4 Crypto IKEv2  SA

Tunnel-id Local Remote fvrf/ivrf Status
1 25.7.73.198/500 25.7.73.193/500 none/none READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify:
PSK
Life/Active Time: 86400/4575 sec

IPv6 Crypto IKEv2  SA

Spoke1#

```

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Conclusión

Se configuró exitosamente DMVPN Fase 3 con IKEv2 y OSPF entre un Hub y dos Spokes. Esta implementación representa la combinación más moderna y escalable de DMVPN: IKEv2 reduce los mensajes de negociación a

4 (vs 6-9 de IKEv1 en P7), mejora el manejo de errores y soporta NAT-T nativo, mientras que la Fase 3 con `ip nhrp redirect` y `ip nhrp shortcut` permite túneles directos Spoke↔Spoke sin intervención continua del Hub. Las dos SAs IKEv2 en cada Spoke (una hacia el Hub y otra directa hacia el Spoke remoto, ambas en estado READY) confirmaron el funcionamiento correcto de la Fase 3. La conectividad end-to-end entre las tres LANs fue verificada exitosamente mediante pings bidireccionales.